

Antenna Datasheet

Product OC: YECT006W1BC
Version: 2.0
Date: 2026-07-20
Status: Released

Product Name: 4G Terminal Mount Rubber Monopole Antenna
Key Features:
Frequency Band: 698–960 MHz, 1710–2690 MHz
Dimensions: 17 mm × Φ 9.4 mm × 43 mm
Efficiency: Up to 76.4 % (EVB)

Overview

YECT006W1BC is a 4G rubber antenna measuring 17 mm × Φ 9.4 mm × 43 mm. This ultra-wide-band 4G antenna provides broad coverage from 698–960 MHz, 1710–2690 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is terminated with SMA Male connector. This low profile, terminal mount omni-directional antenna, ideal for applications where the antenna is required to be discrete, is easy to install with maximum durability assured thanks to its TPE enclosure.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. YECT006W1EC is designed as a monopole antenna, which needs to be mounted on a ground plane to offer high efficiency in all working bands. It is a perfect antenna product for customers that desire highest performance. This high-efficiency, high-gain omni-directional antenna is ideally suited for Zigbee, Bluetooth, IoT Sensors, public safety and security, point of sales terminals, smart home automation, robotics / autonomous.

- **Typical applications include:**

✓ Zigbee
✓ Bluetooth
✓ IoT Sensors
✓ Public Safety and Security
✓ Point of Sales Terminals
✓ Smart Home Automation
✓ Robotics / Autonomous

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

Contents

Overview **1**
Contents **2**

1 Specification **3**
1.1. Electrical 3
1.2. Supported Bands 5
1.3. Mechanical & Environmental 7

2 Drawing **8**

3 Detailed Performance **9**
3.1. S-Parameter Test 9
3.1.1. VSWR 9
3.1.2. Return Loss 10
3.2. Radiation Performance Test 11
3.2.1. Efficiency 11
3.2.2. Average Gain 12
3.2.3. Peak Gain 13
3.2.4. 3D & 2D Radiation Pattern 14

4 Packaging **28**

Contact Us **30**
Legal Notices **31**
Revision History **33**

1 Specification

Test Condition: In Free Space & On 130 mm × 130 mm EVB

1.1. Electrical

Electrical	
Frequency Range	698–960 MHz, 1710–2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional
Antenna Type	Monopole

Electrical – Detail											
Band / SPEC	Band	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR	FS	11.3	5.4	-	12.0	2.6	3.6	6.6	-	-	-
	EVB	7.4	3.9	-	3.8	1.5	1.5	4.0	-	-	-
Max. Return Loss (dB)	FS	-1.5	-3.3	-	-1.5	-6.9	-4.9	-2.7	-	-	-
	EVB	-2.4	-4.5	-	-4.7	-14.4	-13.9	-4.4	-	-	-
AVG Eff. (%)	FS	14.5	29.5	-	28.8	48.7	43.6	28.6	-	-	-
	EVB	38.8	57.4	-	52.4	72.1	70.3	54.8	-	-	-
AVG. AVG Gain (dB)	FS	-8.5	-5.3	-	-6.1	-3.1	-3.6	-5.5	-	-	-
	EVB	-4.3	-2.5	-	-2.8	-1.4	-1.5	-2.7	-	-	-
Max. Peak Gain (dBi)	FS	-2.6 (810)	0.3 (950)	-	1.4 (2170)	2.0 (2320)	1.8 (2410)	0.3 (2510)	-	-	-
	EVB	1.6 (810)	2.3 (860)	-	1.7 (2170)	2.5 (2390)	2.4 (2400)	2.1 (2530)	-	-	-

VSWR	FS	≤ 12.0
	EVB	≤ 7.4
Return Loss	FS	≤ -1.5 dB
	EVB	≤ -2.4 dB
Peak Gain	FS	≤ 2.0 dBi
	EVB	≤ 2.5 dBi

1.2. Supported Bands

5G NR / LTE / LTE-Advanced / WCDMA / HSPA / HSPA+ / GPRS / GSM / NB-IoT				
Band	**Frequency (MHz)**	**Uplink (MHz)**	**Downlink (MHz)**	**Covered**
1	2100	1920–1980	2110–2170	√
2	1900	1850–1910	1930–1990	√
3	1800	1710–1785	1805–1880	√
4	1700	1710–1755	2110–2155	√
5	850	824–849	869–894	√
7	2600	2500–2570	2620–2690	√
8	900	880–915	925–960	√
9	1800	1749.9–1784.9	1844.9–1879.9	√
11	1500	1427.9–1447.9	1475.9–1495.9	-
12	700	699–716	729–746	√
13	700	777–787	746–756	√
14	700	788–798	758–768	√
17	700	704–716	734–746	√
18	850	815–830	860–875	√
19	850	830–845	875–890	√
20	800	832–862	791–821	√
21	1500	1447.9–1462.9	1495.9–1510.9	-
22	3500	3410–3490	3510–3590	-
23	2100	2000–2020	2180–2200	√
24	1600	1626.5–1660.5	1525–1559	-
25	1900	1850–1915	1930–1995	√
26	850	814–849	859–894	√

28	700	703–748	758–803	√
31	450	452.5–457.5	462.5–467.5	-
34	2100	2010–2025		√
38	2600	2570–2620		√
39	1900	1880–1920		√
40	2300	2300–2400		√
41	2500	2496–2690		√
42	3500	3400–3600		-
48	3500	3550–3700		-
66	1700	1710–1780	2110–2200	√
71	600	663–698	617–652	-
74	1500	1427–1470	1475–1518	-
77	3500	3300–4200		-
78	3500	3300–3800		-
79	4500	4400–5000		-

1.3. Mechanical & Environmental

Mechanical	
Antenna Dimensions	17 mm × Φ 9.4 mm × 43 mm
Material & Color	TPE & Black
Connector Type	SMA Male (Gold plating)
Mounting Type	Terminal
Weight	Typ. 5.82 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
RoHS Compliant	Yes

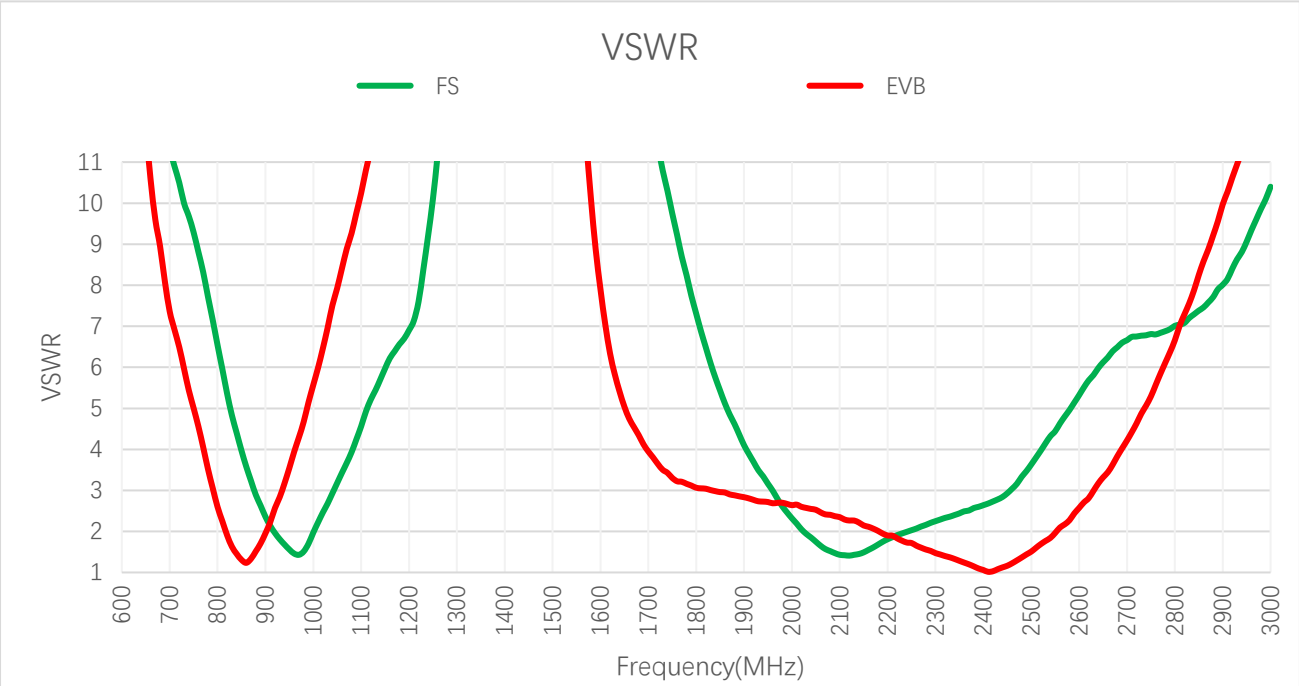
2 Drawing


Hex. 8
17.0±0.5
*43.0±0.5
∅9.4±0.2
SMA Male(Gold plating)
Inside screw inside pin

3 Detailed Performance

3.1. S-Parameter Test

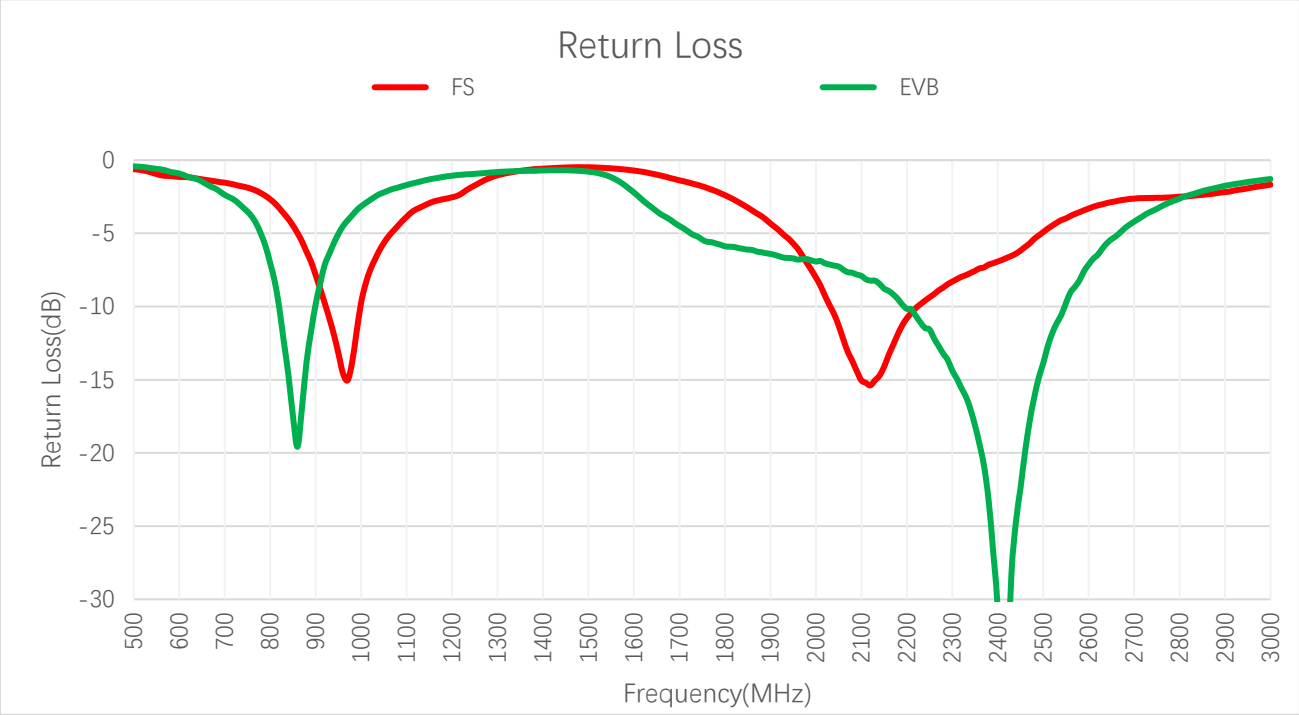
3.1.1. VSWR



VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	10.9	4.8	2.4	1.5	-	12.0	10.3	4.6
EVB	-	-	6.9	1.6	2.0	3.9	-	3.8	3.4	2.9
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	3.2	1.5	2.4	2.9	5.3	6.6	-	-	-	-
EVB	2.7	2.2	1.3	1.2	2.6	4.0	-	-	-	-

3.1.2. Return Loss

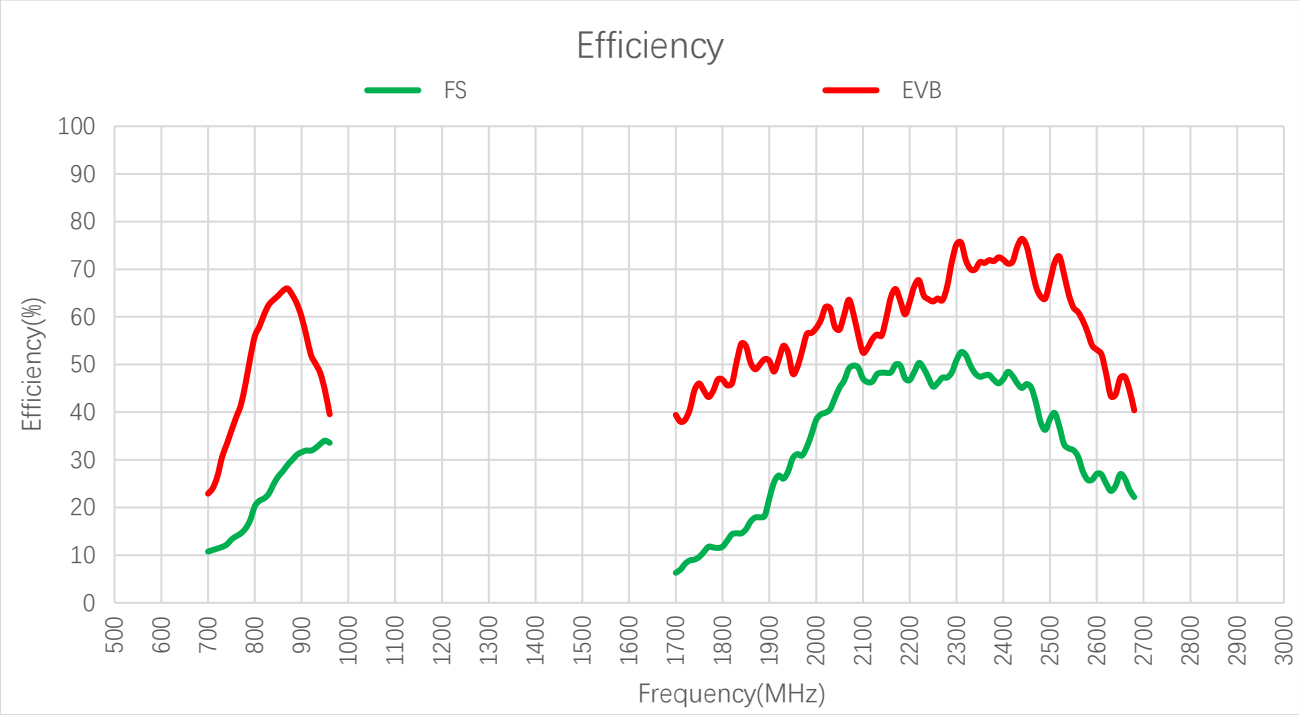


Return Loss (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-1.6	-3.6	-7.9	-14.6	-	-1.5	-1.7	-3.8
EVB	-	-	-2.5	-12.4	-9.8	-4.5	-	-4.7	-5.2	-6.3
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-5.7	-14.7	-7.6	-6.2	-3.3	-2.7	-	-	-	-
EVB	-6.7	-8.4	-18.1	-22.4	-7.1	-4.4	-	-	-	-

3.2. Radiation Performance Test

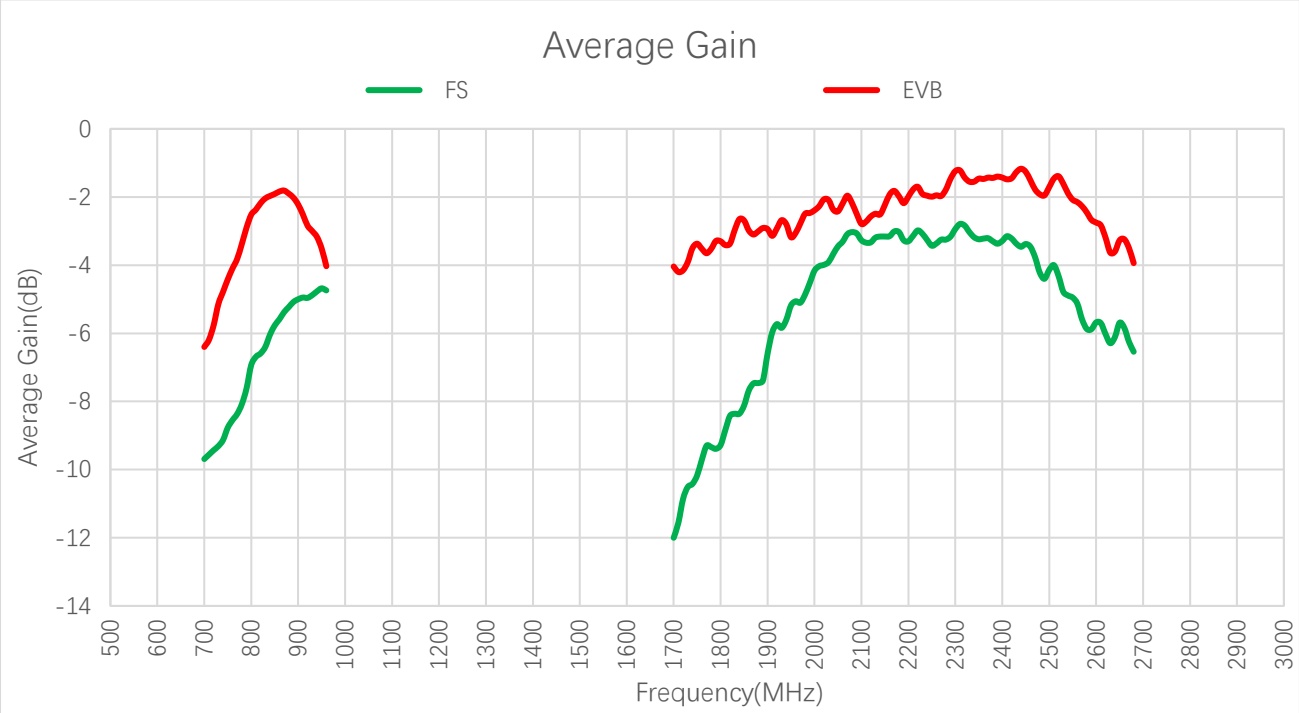
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	11.1	22.9	31.6	33.6	-	7.0	9.1	17.9
EVB	-	-	24.0	62.6	60.0	39.6	-	38.0	44.6	50.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	30.4	48.3	47.4	45.9	27.1	22.5	-	-	-	-
EVB	48.1	56.1	71.5	74.9	53.1	37.9	-	-	-	-

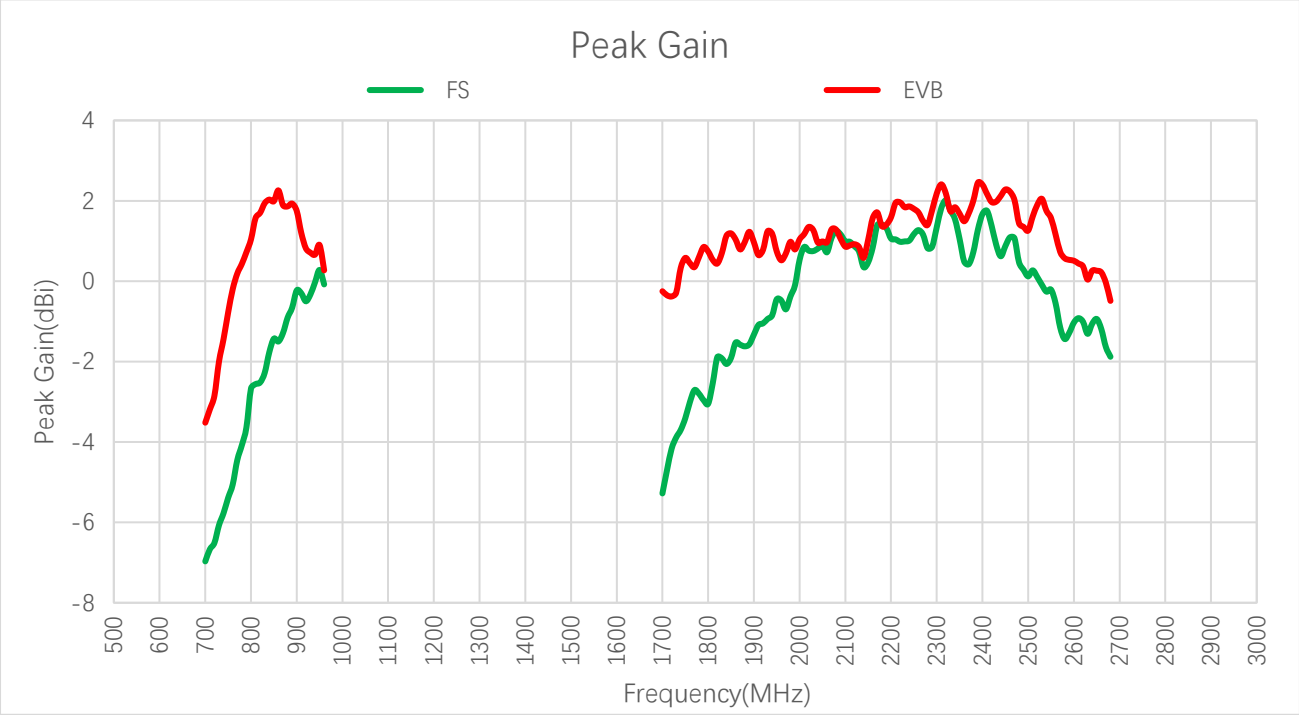
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-9.6	-6.4	-5.0	-4.7	-	-11.6	-10.4	-7.5
EVB	-	-	-6.2	-2.0	-2.2	-4.0	-	-4.2	-3.5	-3.0
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-5.2	-3.2	-3.2	-3.4	-5.7	-6.5	-	-	-	-
EVB	-3.2	-2.5	-1.5	-1.3	-2.8	-4.2	-	-	-	-

3.2.3. Peak Gain

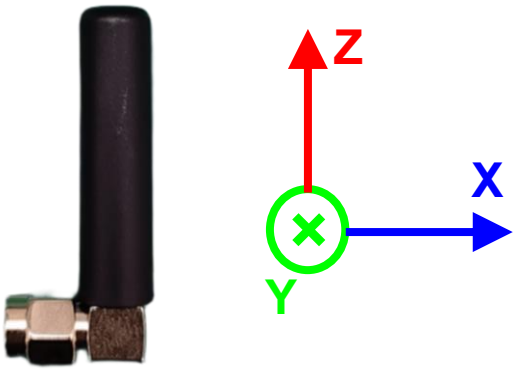


Peak Gain (dBi)

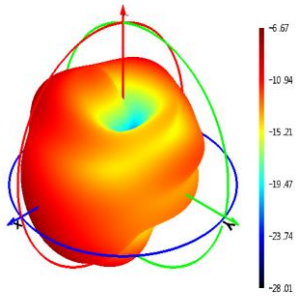
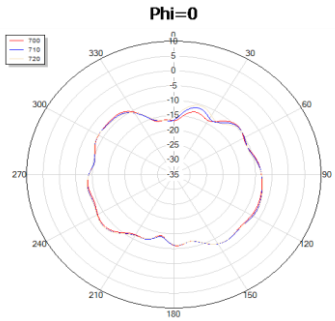
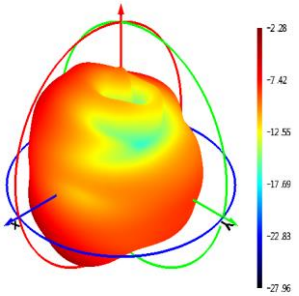
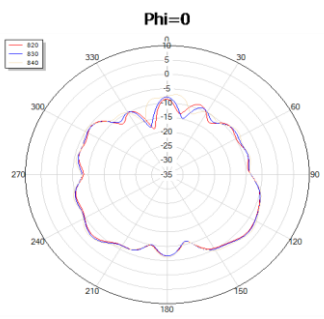
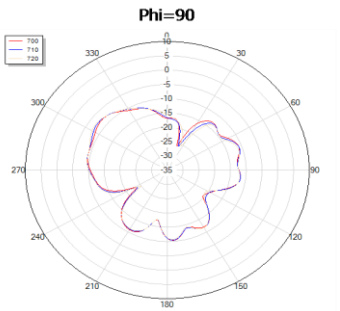
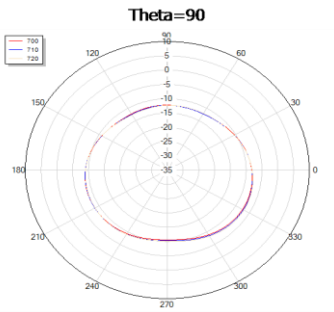
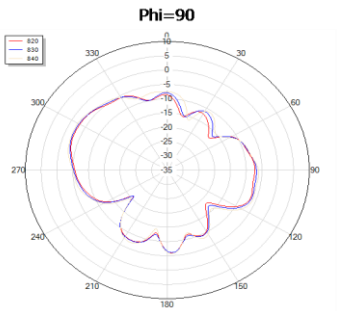
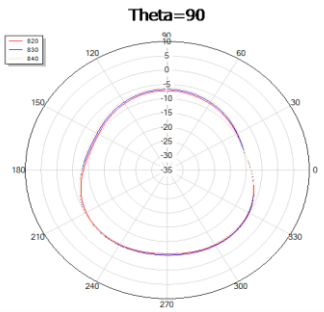
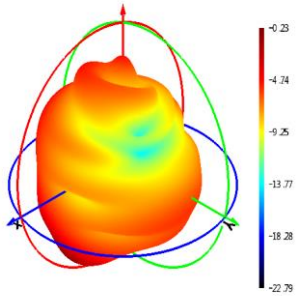
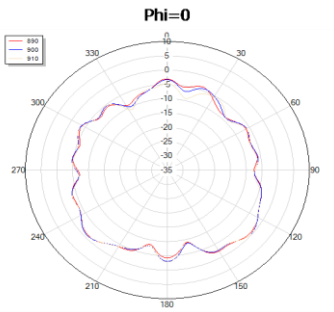
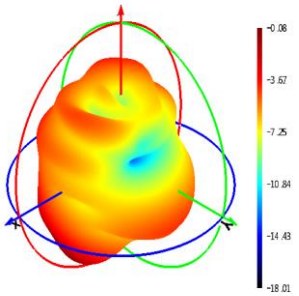
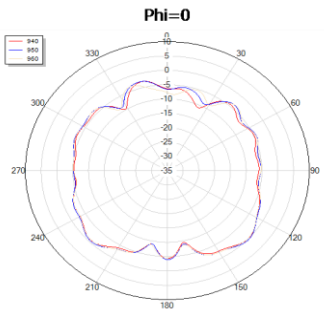
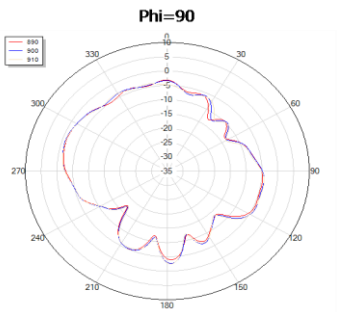
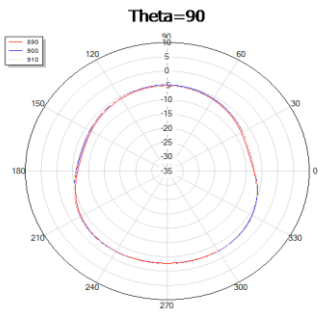
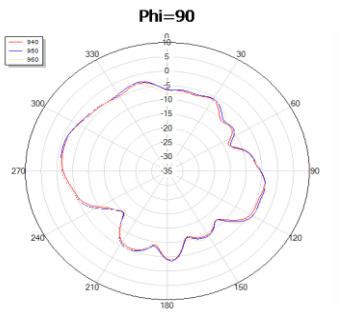
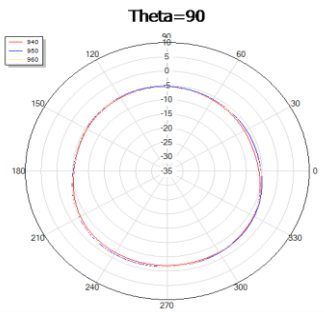
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-6.7	-2.3	-0.2	-0.1	-	-4.7	-3.7	-1.6
EVB	-	-	-3.2	1.9	1.8	0.3	-	-0.4	0.3	1.0
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-0.5	0.4	1.1	0.9	-1.0	-1.6	-	-	-	-
EVB	0.7	0.6	1.7	2.3	0.5	-0.7	-	-	-	-

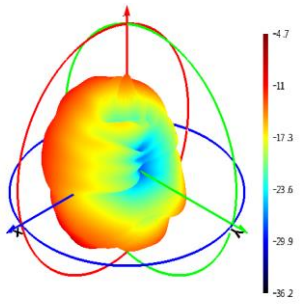
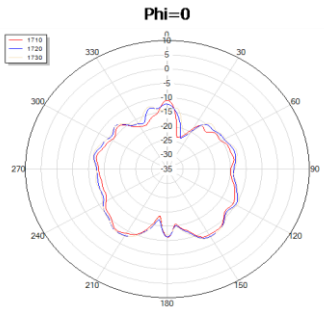
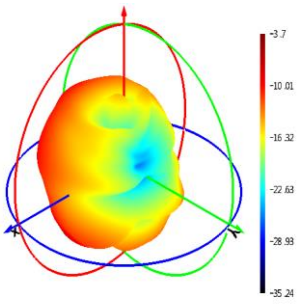
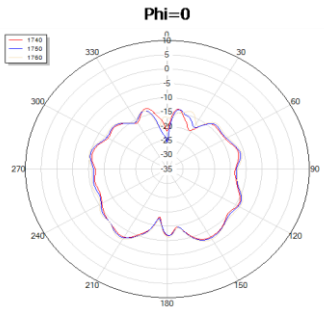
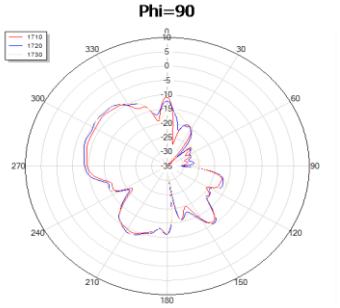
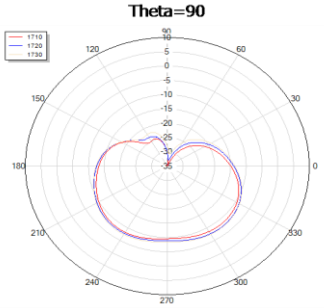
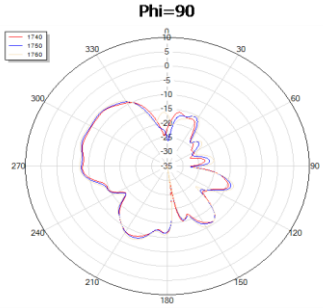
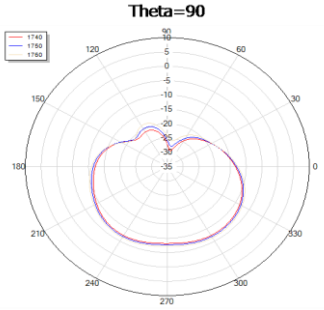
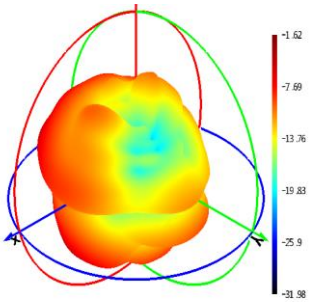
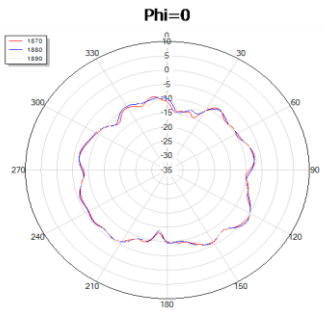
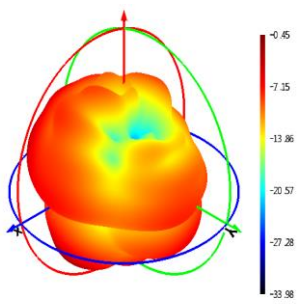
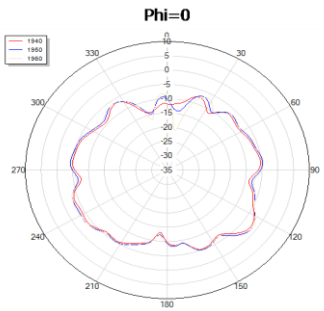
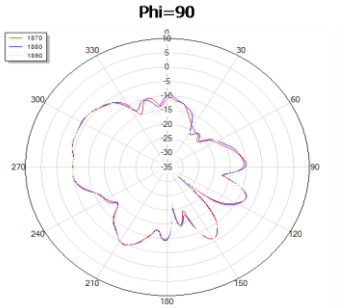
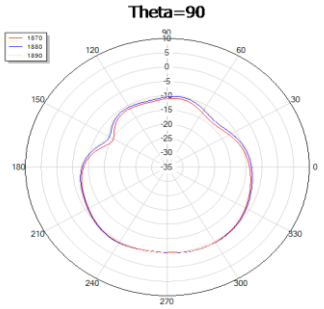
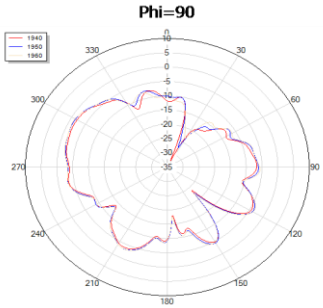
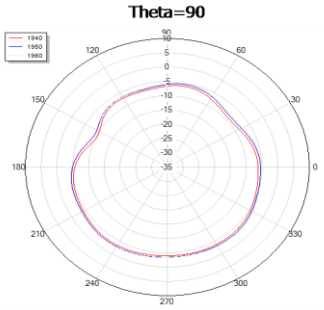
3.2.4. 3D & 2D Radiation Pattern

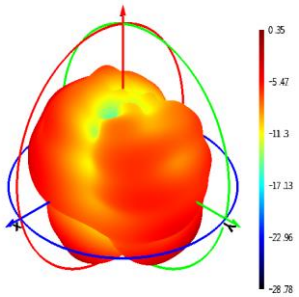
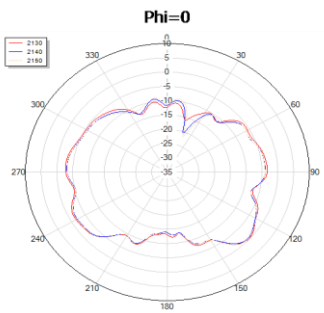
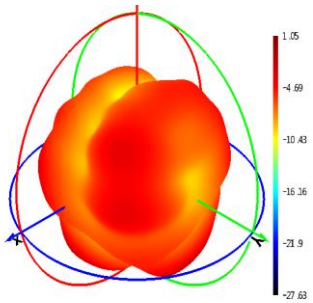
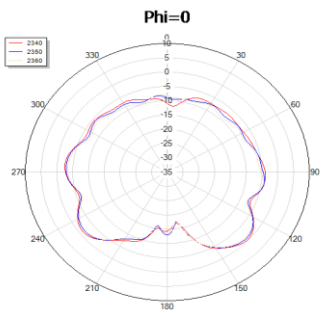
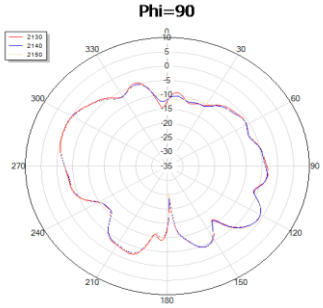
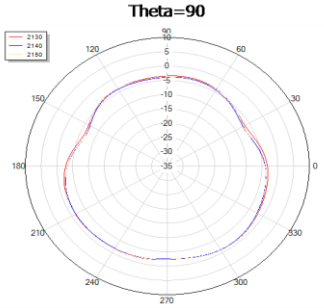
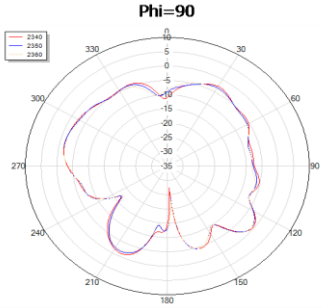
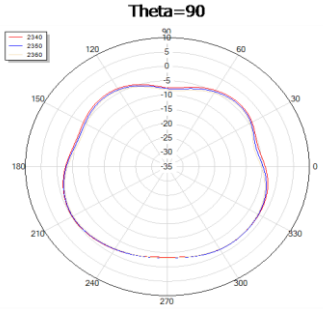
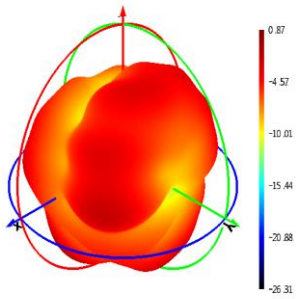
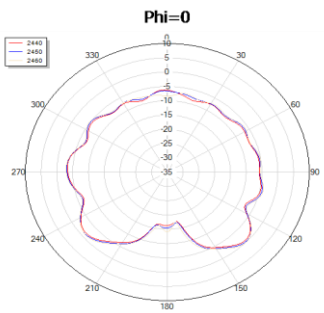
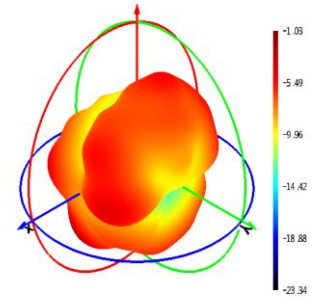
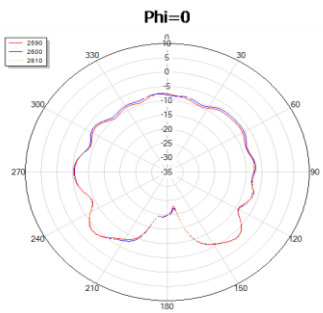
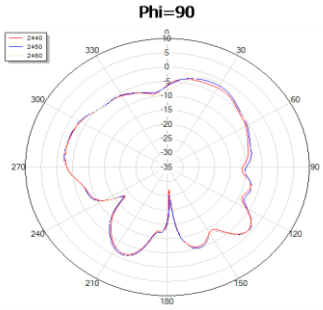
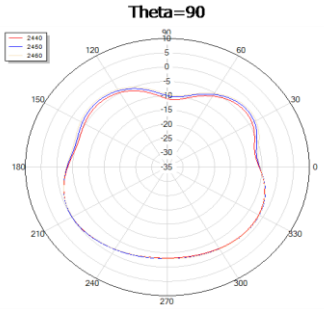
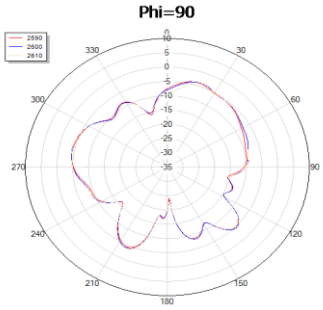
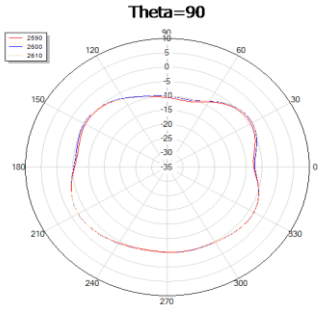
- Test Condition: In Free Space
- Test Chamber: HF-S-1

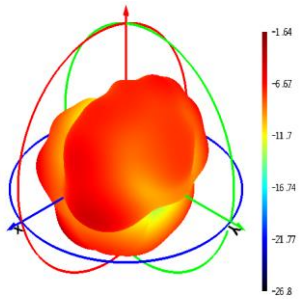
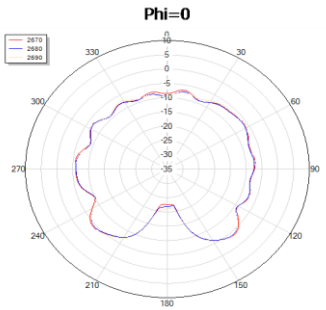
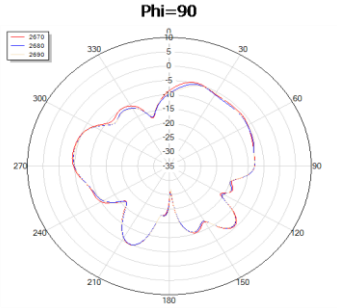
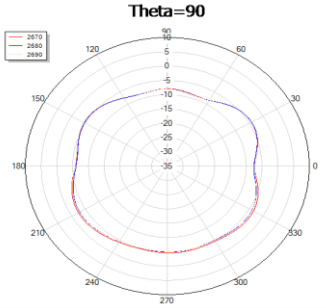


- **FS**

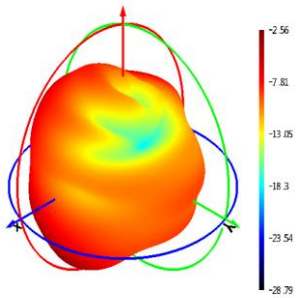
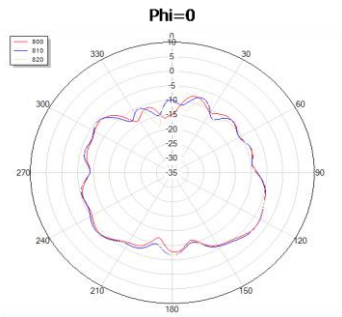
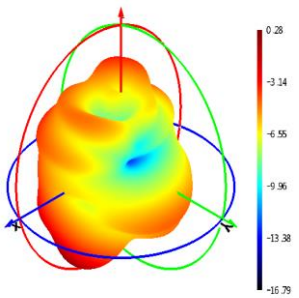
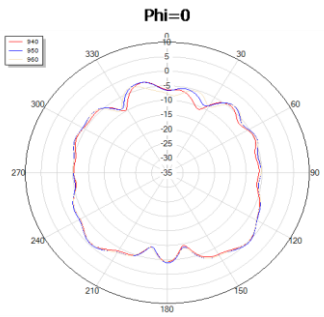
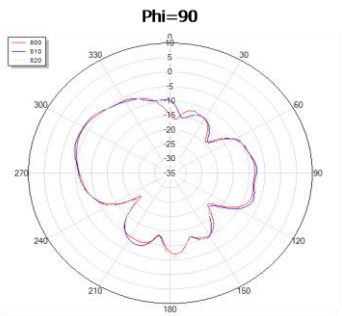
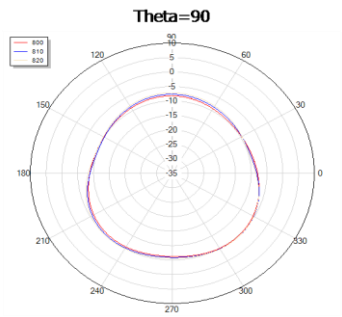
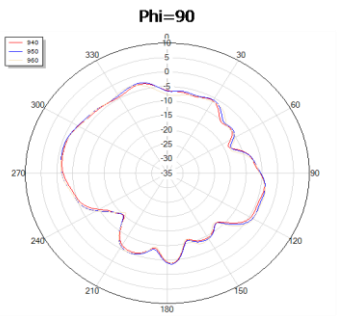
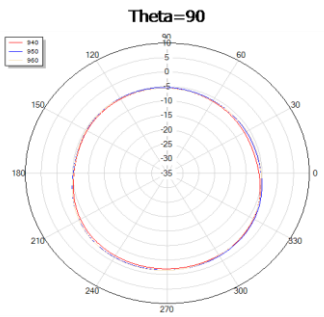
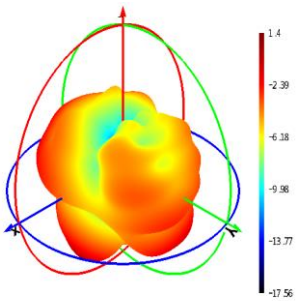
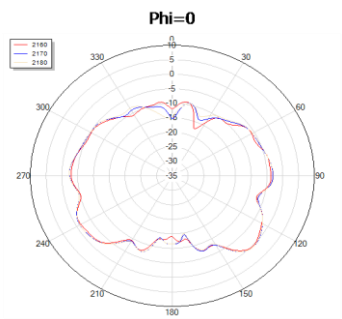
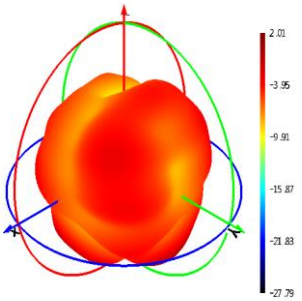
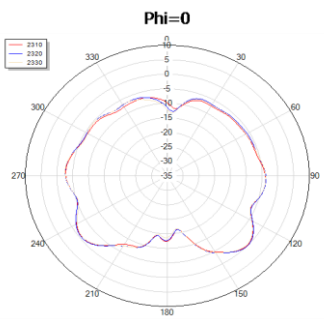
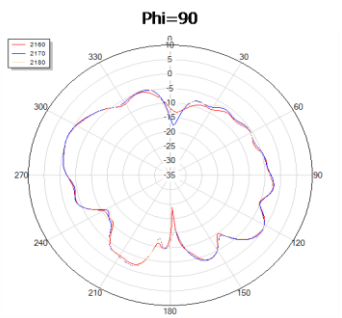
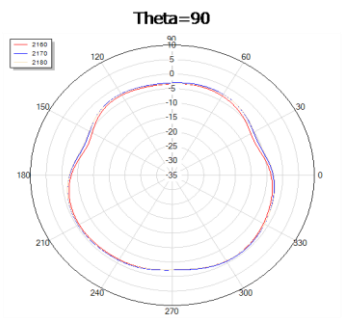
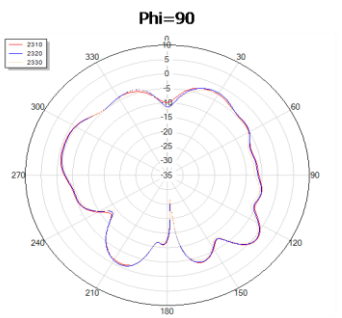
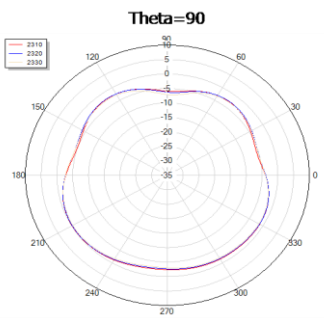
710 MHz		830 MHz	
			
			
900 MHz		960 MHz	
			
			

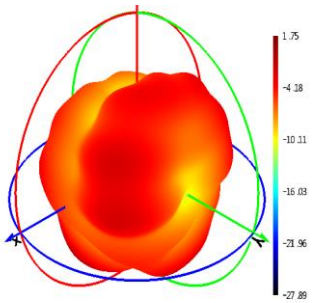
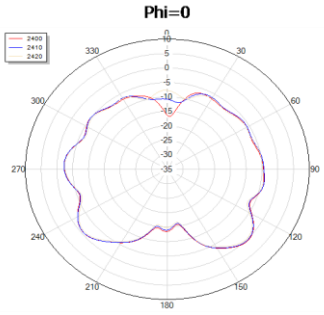
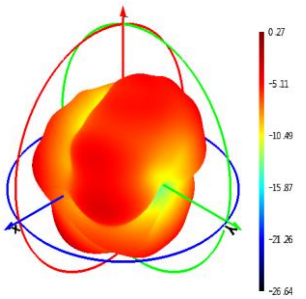
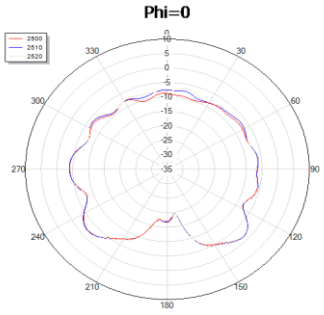
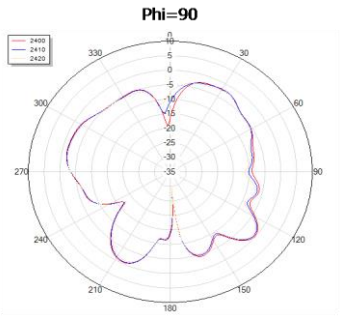
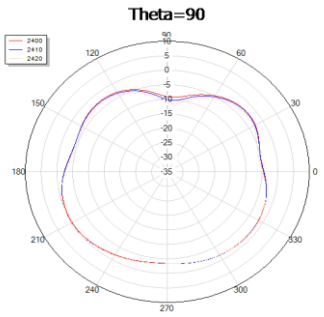
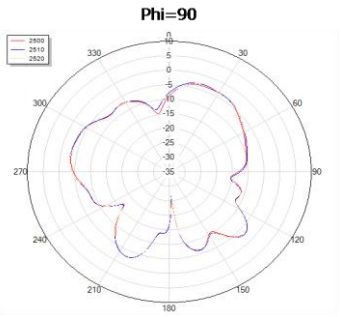
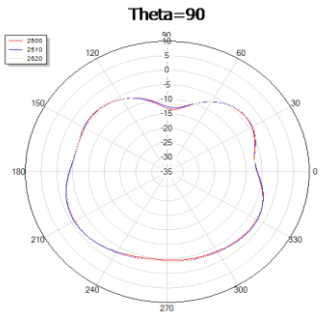
1710 MHz		1740 MHz	
			
			
1880 MHz		1950 MHz	
			
			

2140 MHz		2350 MHz	
			
			
2450 MHz		**2600 MHz**	
			
			

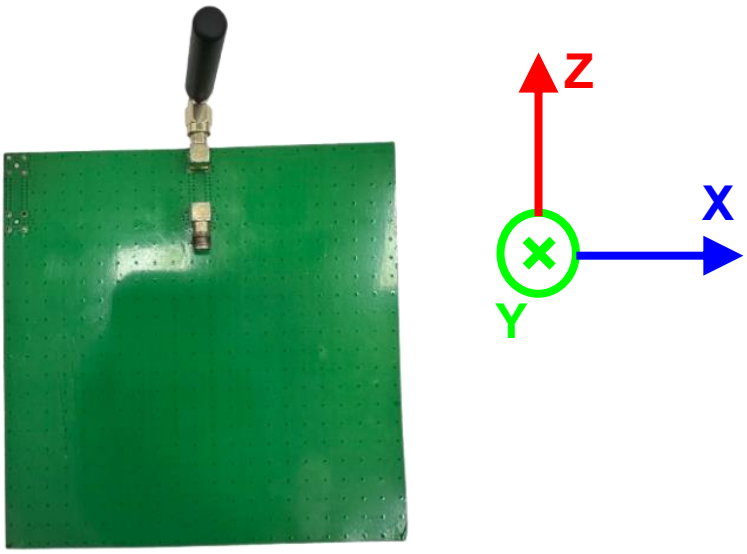
2690 MHz	
	
	

- **Max Peak Gain**

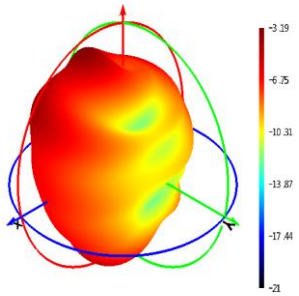
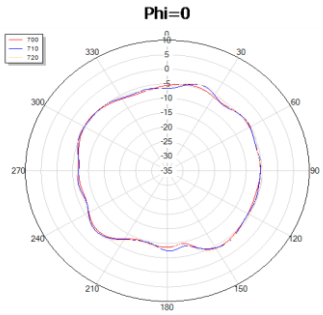
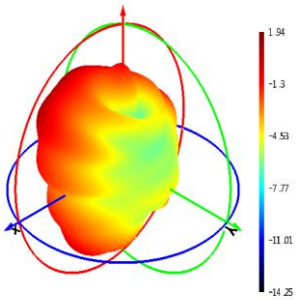
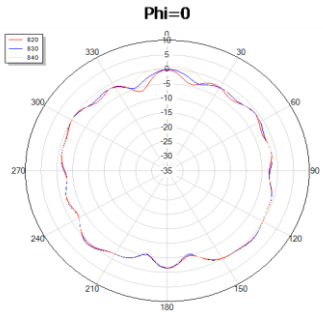
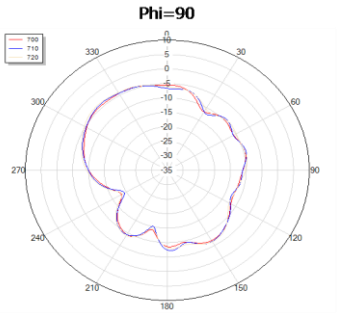
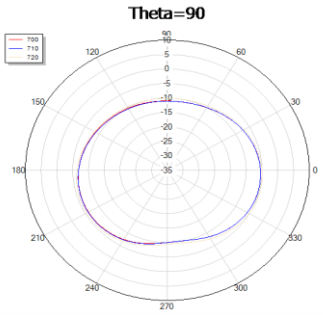
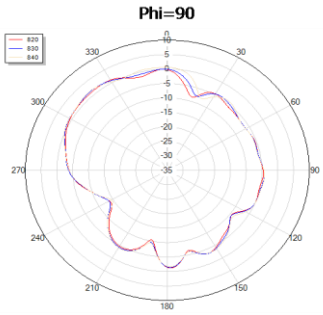
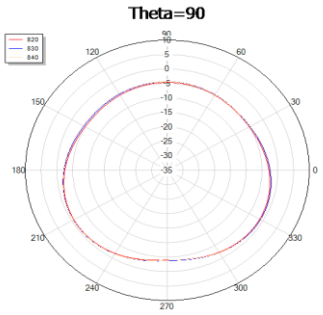
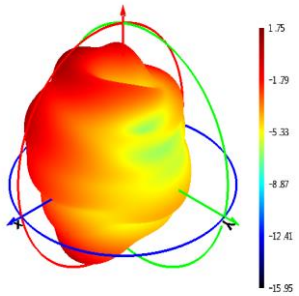
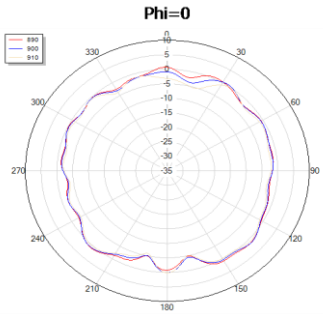
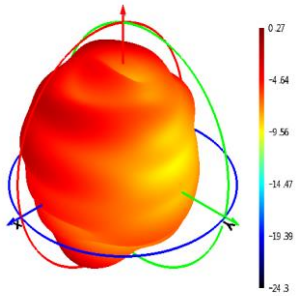
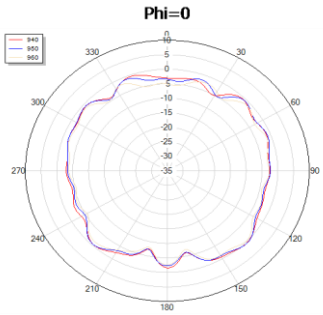
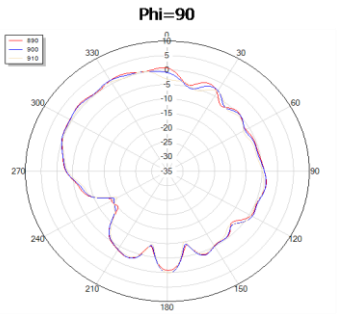
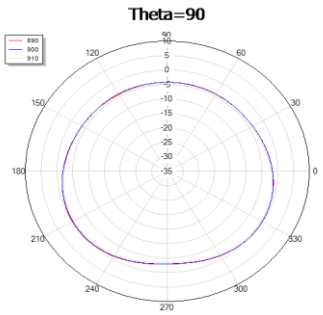
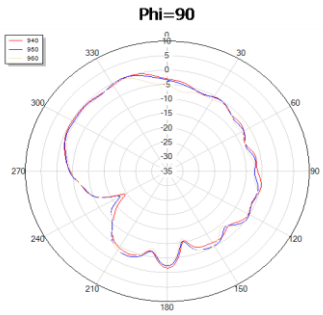
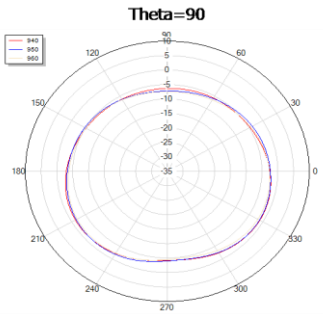
810 MHz		950 MHz	
			
			
2170 MHz		**2320 MHz**	
			
			

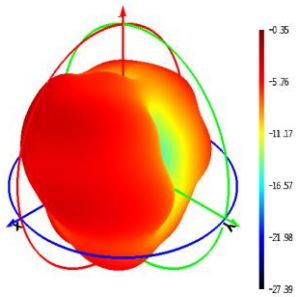
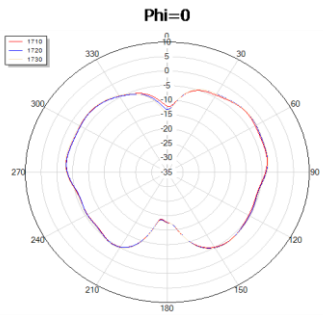
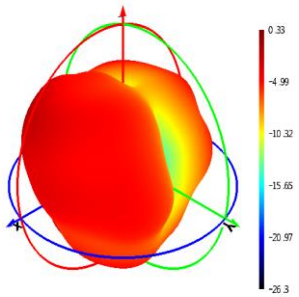
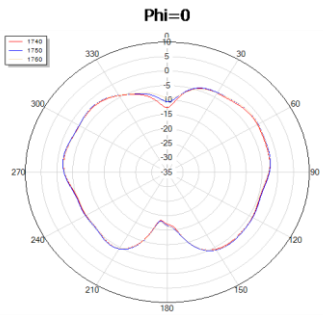
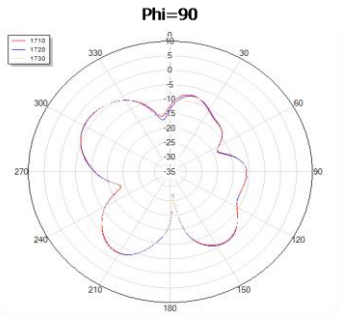
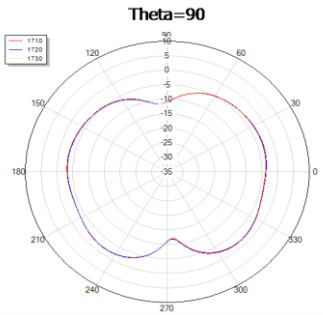
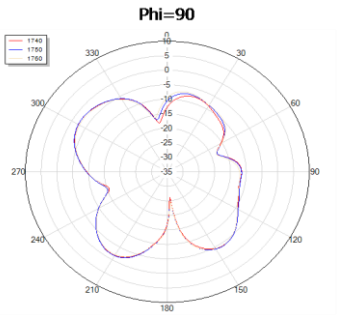
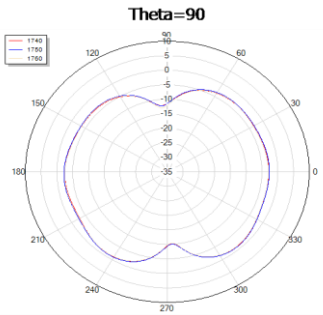
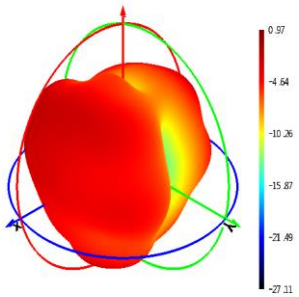
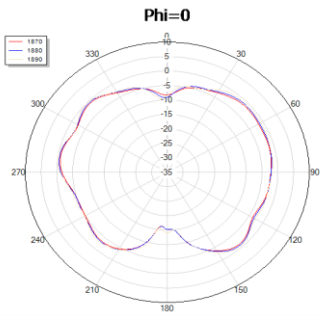
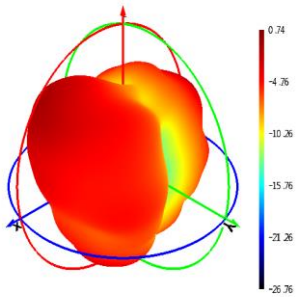
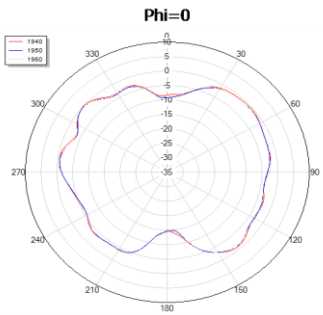
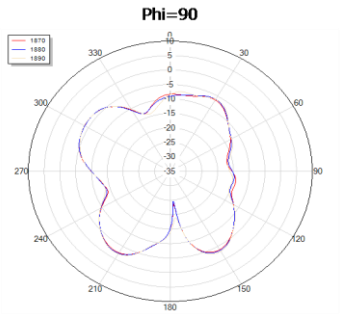
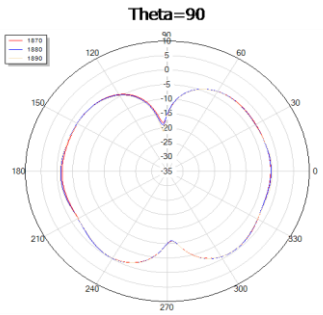
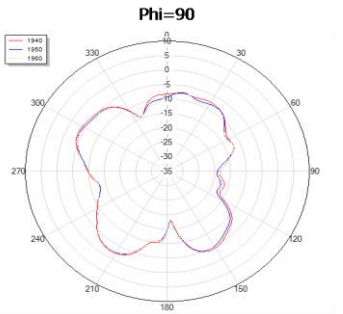
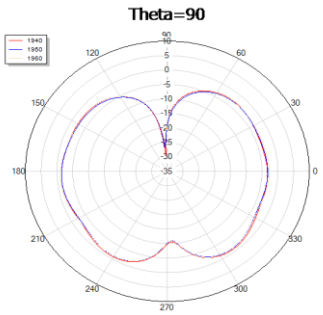
2410 MHz		2510 MHz	
			
			

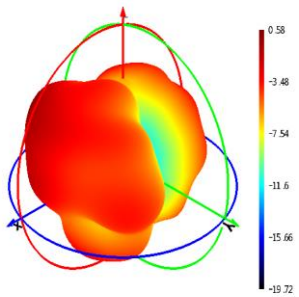
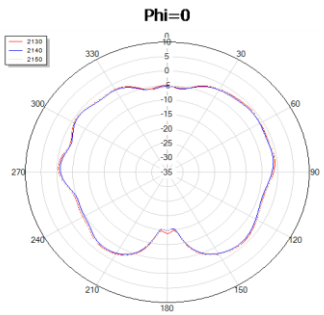
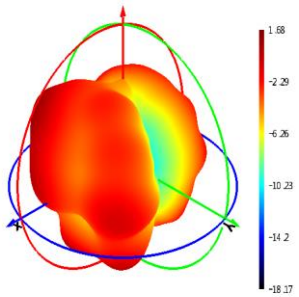
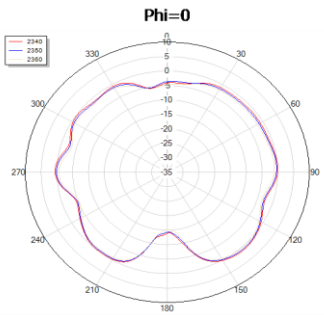
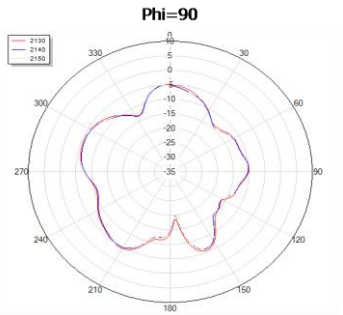
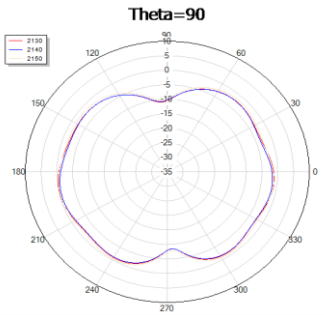
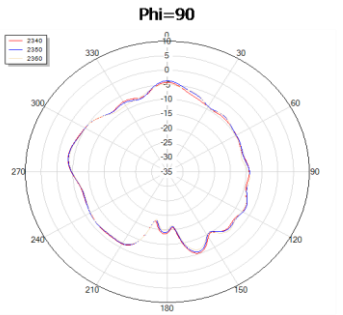
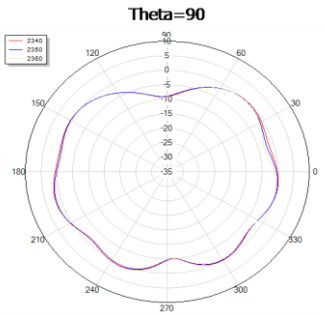
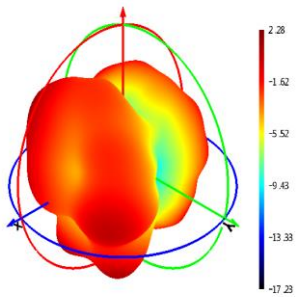
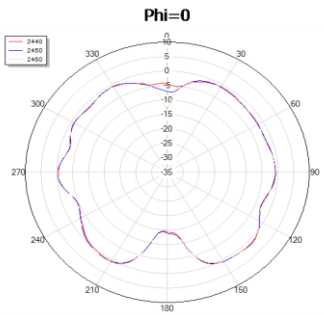
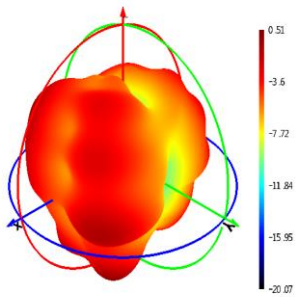
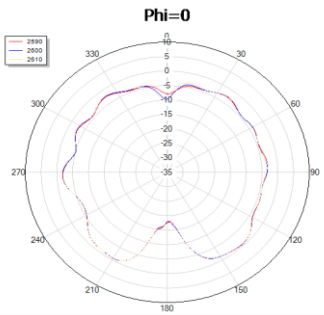
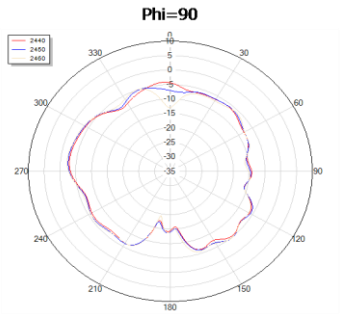
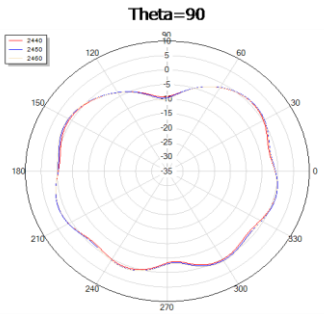
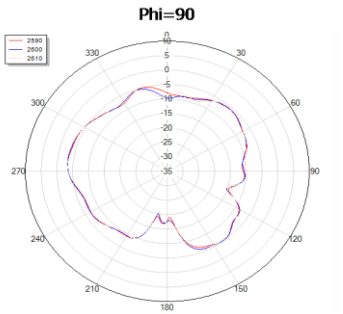
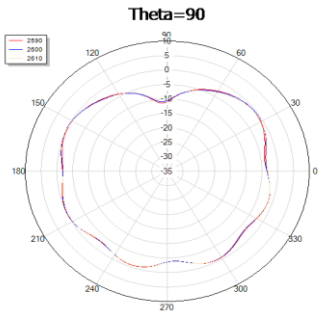
- Test Condition: On 130 mm × 130 mm EVB
- Test Chamber: HF-S-1

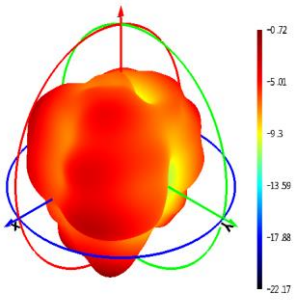
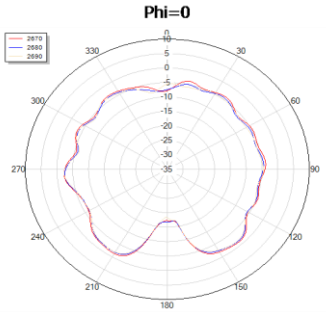
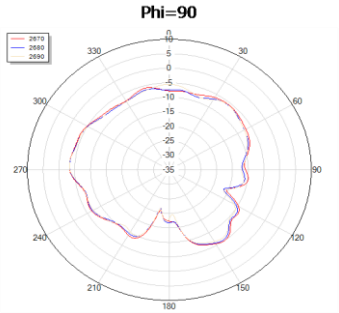
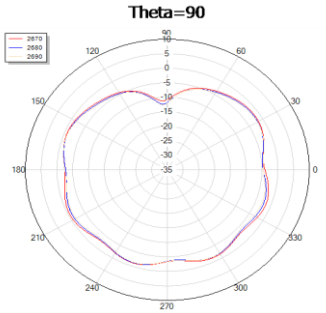


- **EVB**

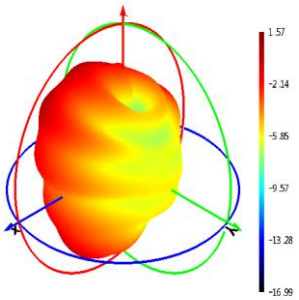
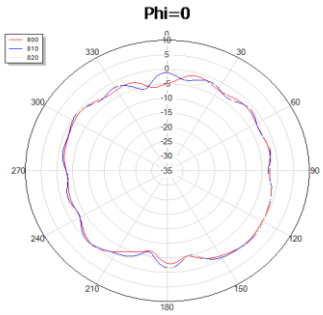
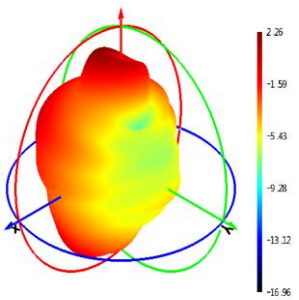
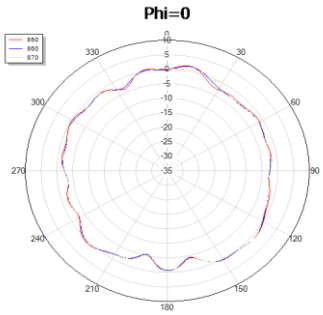
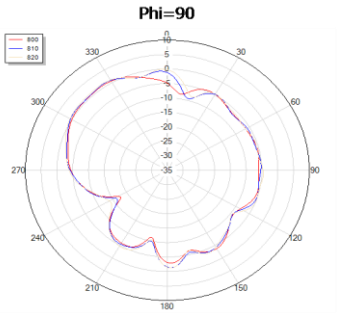
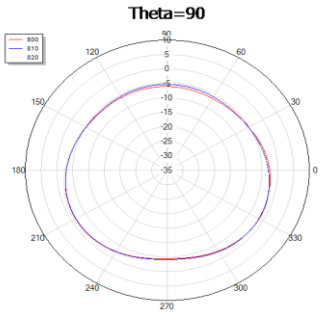
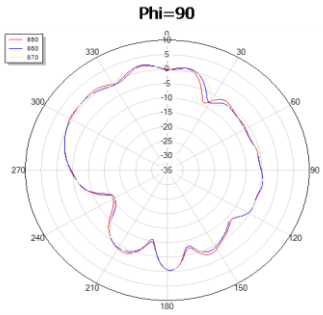
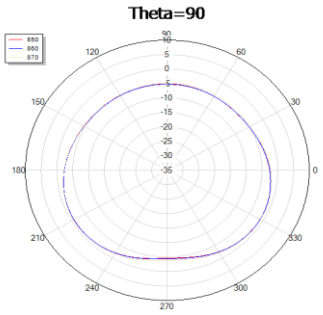
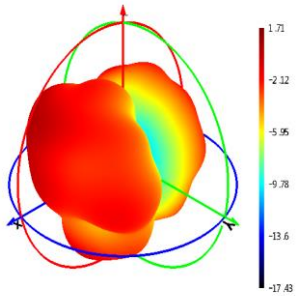
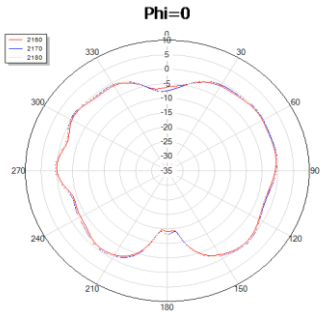
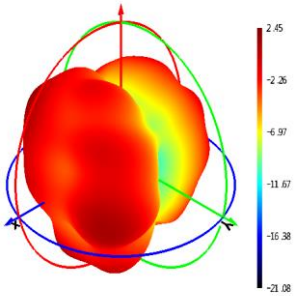
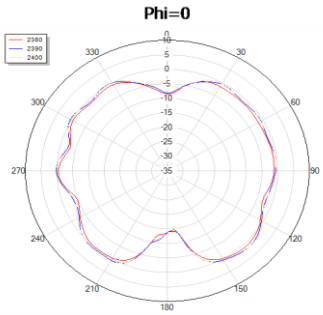
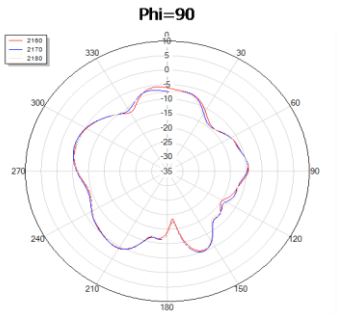
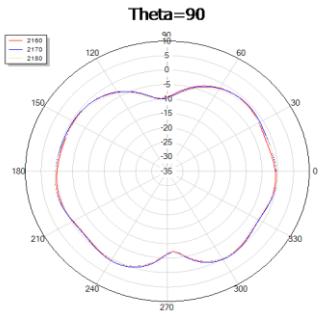
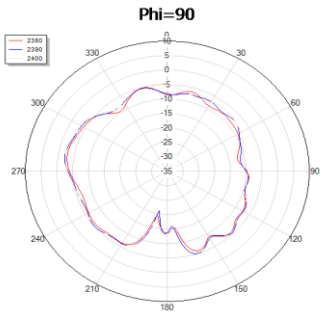
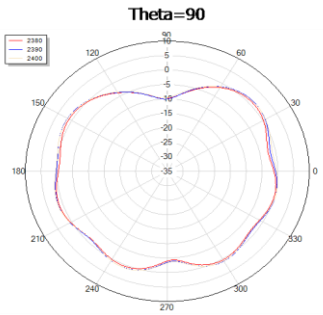
710 MHz		830 MHz	
			
			
900 MHz		960 MHz	
			
			

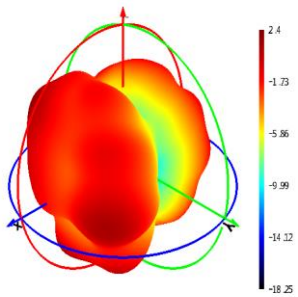
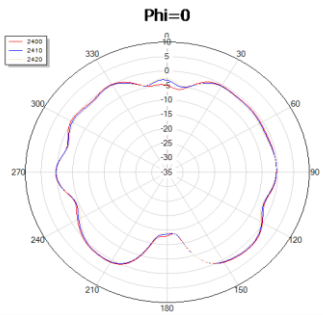
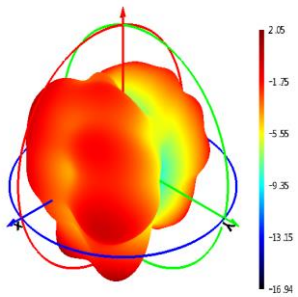
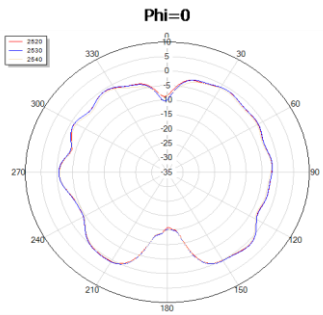
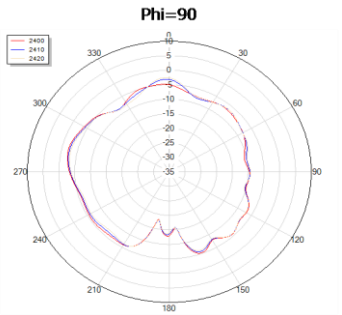
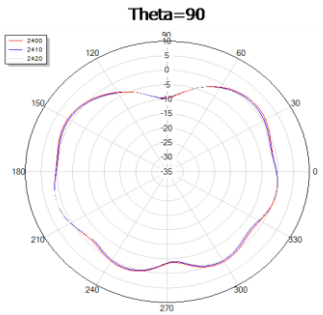
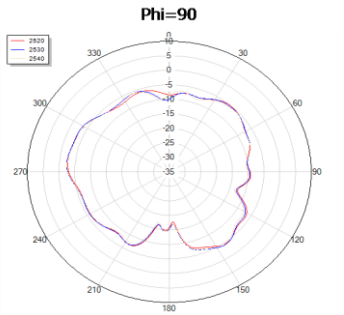
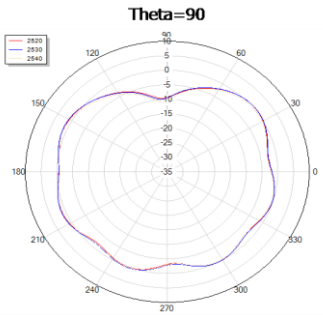
1710 MHz		1740 MHz	
			
			
1880 MHz		**1950 MHz**	
			
			

2140 MHz		2350 MHz	
			
			
2450 MHz		**2600 MHz**	
			
			

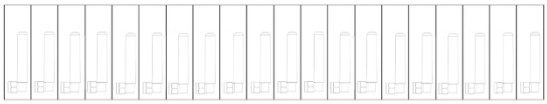
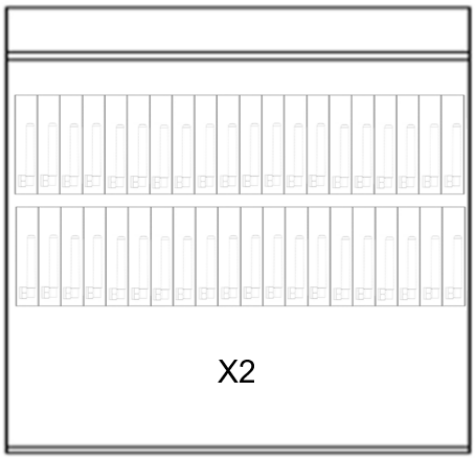
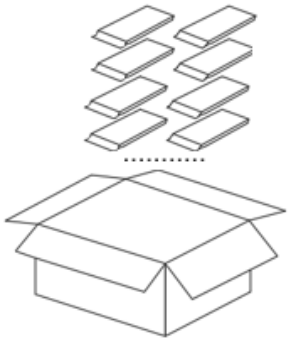
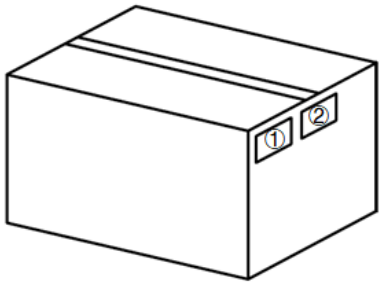
2690 MHz	
 0.72 -5.01 -9.3 -13.59 -17.88 -22.17	 Phi=0 2670 2680 2690
 Phi=90 2670 2680 2690	 Theta=90 2670 2680 2690

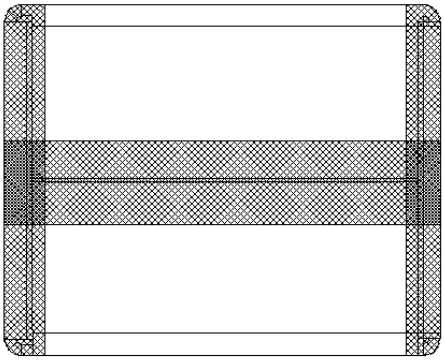
- **Max Peak Gain**

810 MHz		860 MHz	
			
			
2170 MHz		**2390 MHz**	
			
			

2400 MHz		2530 MHz	
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		20 antenna products in a one-piece bag. (20 Antennas / One-piece Bag)
2		40 antenna products in a PE bag. (40 Antennas / PE Bag)
3		(48 PE Bags / Carton Box) (1920 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 470 × 310 × 310 mm
4		**Position for Attaching Labels** ① Carton Label ② Quality Label

5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.
No. 8 Waipojing Road, Sijing Town, Songjiang District, Shanghai 201601, China
Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
https://www.quectel.com/tech-support/.
Or email us at: support@quectel.com.

Legal Notices

We provide this document to support your product design. You are required to design your products based on the specifications and parameters set forth herein. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. You acknowledge and agree that we may add to, amend, or restate this document at any time at our sole discretion without any prior notice to you, and such additions, amendments, or restatements shall be binding upon you.

Use and Disclosure Restrictions

License Agreements

The recipient of any hardware, software, materials, or documentation provided by us shall keep such content confidential, unless expressly authorized by us. The recipient shall not disclose, access, or use any part of the received content for any purpose other than the execution and implementation of the intended project.

Copyright

Our and third-party products hereunder may contain copyrighted materials, including but not limited to protected content, hardware, software, and documentation owned by us or applicable third parties. Unless prior written consent is obtained, you shall not access, use, or disclose any documents or information provided by us, nor shall you copy, reproduce, republish, display, translate, distribute, merge, modify, or create derivative works from any such copyrighted materials. We and the applicable third party retain exclusive rights to all copyrighted materials. No license to any patents, copyrights, trademarks, or service marks shall be granted or transferred. For the avoidance of doubt, no form of purchase shall be construed as granting any license beyond a normal, non-exclusive, royalty-free license to use the product. We reserve the right to pursue legal action against any violation of confidentiality obligations, unauthorized use, or any other unlawful or malicious use of the aforementioned documents and information.

Trademarks

Unless otherwise expressly provided, nothing in this document shall be construed as conferring any rights to use any trademark, trade name, name, abbreviation, or counterfeit thereof owned by us or any third party in advertising, publicity, or any other contexts.

Third-Party Rights

You understand that this document may refer to hardware, software, and/or documentation owned by one or more third parties (“third-party materials”). Use of such third-party materials is subject to all applicable restrictions and obligations set forth herein.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing, course of performance, or usage of trade.

Privacy Policy

To enable product functionality, certain device data may be uploaded to our or third-party servers, including those operated by carriers, chipset suppliers, or servers designated by you. We strictly comply with applicable laws and regulations and will retain, use, disclose, or otherwise process relevant data solely for the purpose of enabling product functionality, or as permitted by applicable laws. Before interacting with any third party regarding data exchange, please be informed of and understand their privacy and data security policies.

Disclaimer

a) We shall not be liable for any damages resulting from failure to comply with applicable operational or design specifications.
b) We shall bear no liability for any inaccuracies or omissions in this document, nor for any damages arising from the use of the information contained herein.
c) While we make every effort to ensure the integrity, accuracy, and timeliness of the features and functions under development, errors or omissions may nevertheless occur. Unless otherwise provided in a valid written agreement, we make no warranties of any kind, express, implied, or statutory, and disclaim all liability for any loss or damage arising from the use of any features or functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage is foreseeable.
d) We assume no legal responsibility for the accessibility, safety, accuracy, availability, legality, or completeness of any information, content, advertising, commercial offers, products, services, or materials on third-party websites or third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2026. All rights reserved.

Revision History

Version	Date	Author	Note
-	2026-04-27	Ruean Liu/ Toby Wang/ Kessie Geng/ Rainey Liao	Creation of the document
1.0	2026-04-27	Ruean Liu/ Toby Wang/ Kessie Geng/ Rainey Liao	First official release
2.0	2026-07-20	Jasper Ma	Updated all test data.



www.quectel.com